The professors in the Mechanical Engineering department at IIT Bombay are currently working on a remarkably diverse and highly specialized range of research topics spanning five major divisions: solid mechanics and design, thermal and fluid engineering, manufacturing processes, control systems and mechatronics, and industrial engineering. Below is an exhaustive and deeply refined list of cutting-edge, active, and recently completed research domains, assembled from their latest publications and research interests. This captures the superset of all specific areas of mechanical research at IIT Bombay, reflecting both their unique technical focus and broader interdisciplinary impact.

## Highly Specialized Research Domains at IIT Bombay Mechanical Engineering

- Heat pump system design for freeze concentration desalination, dedicated outdoor air integration with VRF systems, modeling of low-global-warming-potential refrigerants, solar heating solutions for agro-produce, and solar thermal payback modeling.
- Numerical simulations and direct numerical simulation (DNS) of transitional and turbulent flows, advanced LES (Large Eddy Simulation) modeling, turbulent combustion chemistry tabulation, fluid-structure interaction for biological flows such as obstructed ureters, crystal growth simulations using porous media, and modeling of soot formation in combustion reactors.
- Human-robot interaction modeling, assistive robotics for rehabilitation, plasma science approaches to electrical discharge machining, biostimulation for bioremediation of refinery sludge, 3D laser cladding and repair, magnetic field effect simulation in machining.
- Materials informatics, atomistic simulation of fracture and plasticity in metallic systems, crystal plasticity theory, Bayesian machine learning applications for property inference in complex alloys, finite element implementation for crack mechanics, fatigue life prediction for additive manufactured titanium alloys, multiscale computational mechanics.
- Additive manufacturing of functionally graded and dissimilar materials, in-situ process monitoring for Wire Arc Additive Manufacturing (WAAM), X-ray tomography of material interfaces, advanced joining techniques including friction stir welding and rotary friction welding, anisotropic crack propagation studies in additively manufactured steels, adaptive and hybrid control algorithms for real-time manufacturing automation, metal recycling using friction stir methods.
- Turbulence, rarefied gas flow and slip physics in microfluidic channels, biomimetic device development for isolating biological entities, tangential

momentum accommodation for rarefied gas, phase change studies in micro-scale devices, impinging jets, and complex heat transfer analysis.

- Continuum mechanics, multiscale statistical mechanics, heat conduction in polymers and graphene, fatigue and ballistic impact modeling, finite element methods for multiscale transport and fracture, microrheology in polymers and composite materials.
- Powder bed fusion additive manufacturing for metals/alloys, high-speed joining through gas metal arc brazing, numerical heat and fluid modeling of welding and joining, real-time heat input control for robust manufacturing, advanced control of current/voltage in brazing, distributed manufacturing system analytics.
- Shock-induced material amplification and transmission reduction in aluminum foams, laser surface remelting and micro-hardening in superalloys, surface hardening via oxygen diffusion in niobium, high-temperature fracture and fatigue analysis in advanced materials.
- Kinematics of tensegrity and robotics, industrial artificial intelligence and machine learning applications, modeling of human locomotion, design of safety assist devices, human motion simulation, remotely operated vehicles, simulation for impact-resistant structure optimization, advanced image analysis for industrial defect detection, structural health monitoring using AI.
- Atomistic simulations of heat and electrical transport in next-generation materials for energy applications, high-throughput ML calculations for material discovery, thermochemical and electrochemical reduction for fuel synthesis (methanol, ammonia), development of workflows for fatigue and durability evaluation of additive manufactured metals.
- Combustion visualization for propellants and energetic materials, advanced diagnostics for homogeneous charge compression ignition engines, droplet combustion for blended fuels, fire dynamics and suppression device characterization, emission reduction through innovative fuel injection.
- Heat transfer in nuclear safety, two-phase flow modeling, CFD-based pipe wall corrosion prediction, stochastic modeling for nuclear reactor degradation, boiling in microchannels, chill down phenomena in cryogenics and liquid nitrogen systems.
- Deep learning and big data analytics for smart manufacturing, optimization of fibre-reinforced composite pressure vessel strength, Internet of Things (IoT) product development, advanced morphological and crystallographic 3D characterization, image analysis for manufacturing, robust AI for segmentation and damage prediction, algorithmic control in additive manufacturing.
- Ultra-precision machining processes, LIGA and micro-nanomachining, advanced grinding and development of chip geometry models for superalloys,

finite element-based vibratory response prediction, time-domain vibratory deflection modeling during flank milling.

- Multi-phase flow and latent heat thermal energy storage, renewable energy integration with PCM for solar applications, quasi-steady state modeling for two-phase organic Rankine cycle heat exchangers, encapsulated phase-change materials for thermal management.
- Computational multi-fluid dynamics, advanced CFD for fish-like hydrofoils, force coefficient and vortex dynamics for engineered objects, Printed Circuit Heat Exchanger thermal optimization, fluid-structure hydrodynamic models for rapid product development.
- Acoustics and vibration modeling in porous multifunctional materials, elastic and acoustic response prediction for periodic foams, thermal-acoustic co-design for noise-reducing heat sinks, reverse-engineering cochlear amplification, parametric design and experimental validation for acoustic materials.
- CFD of turbulent combustion in engines, soot modeling and thermophoretic deposition for engine optimization, dual-fuel system exergy analysis, flame temperature measurements using novel pyrometry, hydrokinetic turbine design, jet flame impingement heat transfer.
- Metal casting and process simulation, Bayesian inference for defect analysis in industrial investment castings, foundry data analytics, prosthesis design using finite element analysis, distributed manufacturing opportunities via advanced analytics.
- Micronano manufacturing: ultra-short pulsed laser ablation, electroless plating post-laser processing, electric discharge and electrochemical machining, multiphysics modeling for rapid manufacturing, improvement of polymer surface properties for subsequent manufacturing steps.
- Non-Fourier heat transfer in nanomaterials, sub-K heat exchanger design, ultrafast phase transition dynamics in advanced battery and quantum materials via optical and X-ray spectroscopy, probing of atomic and electron spin behaviors, optimal stress wave design for stability in composite structures.
- Reliability engineering for digital twins and manufacturing analytics, spare parts forecasting, shop floor control indices, goal programming for warranty optimization, value chain framework for manufacturing efficiency.
- Robotics, mechatronics, dynamics, and real-time control, teleoperated linkage simulation under communication delays, impulsive force generation by aerial vehicles, motion planning for biomechanical maneuvers, contact force tracking algorithms for planar manipulators.
- Vibration diagnostics, guided-wave based structural health monitoring, deep learning for fault analysis in rotating machinery, dimension theory

applications for bearing clearance, fiber-reinforced composite damage analysis via simulation and experimental methods.
- CAD/CAM geometric reasoning, feature-based manufacturing process modeling, AI/machine learning applications for process optimization in design and manufacturing, feature-based modeling for industrial automation.
- Fracture mechanics, computational solid mechanics, crystal plasticity and wave propagation modeling, multiscale simulation of deformation and bulk metallic glasses, shape memory alloy failure and glass transition behavior, fluid-structure interaction simulations.
- Advanced rapid manufacturing automation with CNC, computer graphics applications in CAM, rotary-wing micro-aerial vehicle design, helicopter tail rotor elimination via rotary-wing engineering, ablation process modeling in rapid prototyping, substrate tilting optimization for additive manufacturing.
- Biomedical applications: bioheat transfer modeling in laser-induced tissue heating, 4D CT analysis for joint replacement stability, frugal biomechanics and mechanical design of orthopaedic devices, AI for kinematics in joint and ligament simulation, in-vitro testing and gait simulator development.
- Geophysical and atmospheric flow modeling, stratified buoyancy-driven mixing studies, double-diffusive convection, fluid instability via laser instrumentation, spray jet numerical analysis, magnetohydrodynamics in liquid metal batteries and engines, internal wave propagation modeling, global potential energy surface studies in molecular collisions.

This superset represents not just broad domains, but highly granular research topics—from machine learning in material property inference and advanced multiscale simulation workflows for fracture, to the design and experimental validation of acoustic materials and biomedical devices. Students collaborating with these faculty members engage in the full spectrum of experimental, computational, and applied engineering research, contributing directly to advances in manufacturing, energy systems, health care, materials science, and computational modeling.